John Blake Caven

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I build GPU rendering and real-time software, measure and de-risk new techniques, then turn them into tools other people can use. I am especially interested in physically based light transport and accurately simulating the real world.
I recently completed my MS in Computer Science at the University of Notre Dame, focused on machine learning and real-time simulation. At Notre Dame Studios, I pitched and built real-time virtual production capabilities using Unreal Engine 5, stYpe tracking, custom C++ integrations, real-time depth and pose approximation, and digital-twin-based workflows.

EDUCATION

University of Notre Dame | Notre Dame, IN

Master of Science, May 2026; Bachelor of Science, May 2025

Major: Computer Science *Dean's list* Spring 2025

TECHNICAL SKILLS

C++, Slang, Vulkan, Nsight Graphics, GLSL, OpenGL, SDL3, PyTorch, LibTorch, Unreal Engine, Godot Engine, OpenCV, Blackmagic DeckLink SDK, Python, Docker, AWS, Blender, Linux, MicroPython

SELECTED PROJECTS

Offline Monte Carlo Path Tracer | BCaven/OfflineRaytracingGPU August 2026 – present

- Progressive spectral GPU Monte Carlo path tracer in Slang and C++ using Vulkan. Supports triangle, sphere, and Gaussian splat primitives. Experimental mixed internal nodes allow for nested instancing and self-repeating geometry.
- Custom compute pipeline supports 8-wide k-DOP, binary AABB, and stacked transforms in the same structure.
- Renders 8 billion instances of Suzanne using 2 GB at 43.38 ms per sample pass on an RTX 4090 at 1280x720.
- Interactive camera for previewing scenes before rendering final images using temporal accumulation.
- Nsight Graphics showed L1TEX (LGSB) latency caused 98% of warp stalls. Nested instancing cut that to 57%.

Offline Raytracing CPU | BCaven/OfflineRaytracingCPU June 2026 – July 2026

- Monte Carlo Path Tracer in C++. Extends the *RayTracing in One Weekend* series with triangle meshes, Gaussian splats, instanced BVHs, and chunk-based multi-core rendering. This project is the CPU precursor to my GPU path tracer.

C++ Game Engine | BCaven/GameEngine Spring 2026

- From scratch game engine in C++ and GLSL. Supports texture and object instancing and multi-threaded game loops.
- Native CUDA and LibTorch support, designed to provide sandbox for experimental machine learning techniques.

PROFESSIONAL EXPERIENCE

GESS Lab | Notre Dame, IN January 2025 – May 2026

Research Assistant

- Designed and implemented a real-time particle simulation in C++ inside texture buffers to simulate aeolian patterns. 20x performance increase over comparable method.
- Restructured GLSL shaders to visualize particle simulation on GPU for a more efficient rendering pipeline with no unnecessary CPU cost and higher fidelity graphics.
- Comprehensive literature review on modern computer graphics and physics simulations.
- GESS Lab presented at the first Great Lakes Graphics Workshop at the University of Chicago.

Notre Dame Studios | Notre Dame, IN May 2025 – May 2026

Lead Software Engineer

- Pitched and built on-set virtual production studio from scratch using UE5. Periodically presented live demonstrations of the system to the Assistant Vice President for Academic and Community Technology Experience at Notre Dame.
- Production-tested Gaussian splat pipeline: 5 million splats at 59.94 fps in UE5 with live video compositing.
- Implemented real-time machine learning pipeline using UE5 NNE with a custom blueprint wrapper library.
- Designed and built a real-time machine learning pipeline using Blackmagic Design's Decklink and Libtorch to offload heavier models with a 3.34x performance improvement over native UE5 NNE ONNX runtime.
- Relightable digital twin integrating real footage into virtual scenes using per pixel estimated depth.
- Led small team that built and shipped version-controlled tools: aligning arbitrary numbers of child actors to splines, Maya to UE5 importing with texture support, and polymorphic interactive elements.
- Designed, drafted, and presented proposal for interdisciplinary collaboration between Computer Science and Engineering, Film, and Notre Dame Studios for future research, production, and student learning opportunities.

CDW-G | Notre Dame, IN August 2024 – April 2026
On-Campus Intern

- Provided subject matter expertise and on-campus technical sales support as part of CDW-G's account team serving the University of Notre Dame.
- Developed relationships across three campus partners to understand wants, needs, and solutions to meet growing AR needs, presented information to company leadership team to create custom solutions for departments at Notre Dame.

TEACHING, TALKS, AND LEADERSHIP

Art and Science of Human Flourishing | Notre Dame, IN January 2026 – May 2026
Graduate Teaching Assistant

- 90-minute guest lecture on *Students' Relationship with Technology* to a mixed group of students, faculty, and staff.

Advanced Game Development | Notre Dame, IN December 2025 – May 2026
Graduate Teaching Assistant

- Supported five undergraduates and one graduate student when building their own engines.
- Taught lecture on introduction to deferred rendering.
- Self-taught course material in the month leading up to the start of the class to better support students.

Game Development | Notre Dame, IN August 2025 – December 2025
Graduate Teaching Assistant

- Supported 20 undergraduates through office hours, problem solving and custom solutions using Godot Engine.
- Wrote a batch download and extraction script using GraphQL and Python to expedite the grading process.
- Built examples of common components (managing multiple cameras, inventory systems, splitscreen multiplayer) in Godot Engine to serve as inspiration for students' semester projects.
- Coordinated with the Video Game Club to host playtesting sessions for students in Game Development.
- *Alternative Uses of Game Engines*: Highlighted emerging uses of game engines such as Virtual Production and XR. Used discussion and open-ended questions to encourage participation and creativity when students built their own projects.
- *Best Practices in Godot*: Discussed key differences between Godot and UE5 and the best practices therein.

Technical Concepts of VFX II | Notre Dame, IN January 2025 – May 2025
Teaching Assistant

- Supported 25 undergraduates through teaching Houdini, DaVinci Resolve, and Blender 3D, focusing on simulations and compositing for offline-rendered visual effects.
- Independently designed and implemented batch offline rendering solution for Houdini in the Center for Research Computing to allow up to 500 students to utilize university compute clusters when rendering large 3D scenes.
- Coordinated a team of four TAs offering one-on-one advising, weekly office hours, and custom solutions.

Innovate-o-thon | Notre Dame, IN Fall 2021, Spring 2022

- *Top Prize Winner*: Led five-person team to the **top prize of $2,000** in the Fall 2021 and Spring 2022 competitions.

ADDITIONAL PROJECTS

Haptic Hat | BCaven/haptic-hat December 2024 – January 2025

- Embedded sensory-feedback device: MicroPython on a Raspberry Pi Pico translates gyroscope and compass (MPU9250) data into vibration patterns through motors sewn into a hat brim, with custom motor PWM control.

Creativity Optional | BCaven/creativity-optional Spring 2024

- Real-time Streaming Processor: lead developer for OSS that allows the user to send arbitrary real-time data to a custom 3D scene. Tested at 60 fps with real-time audio and system information streams.
- Cross-platform support for Mac, Windows, and Linux using Flask, ThreeJS, and Docker containers.

Elgato Light Controller | BCaven/elgato-light-controller December 2023

- Reverse engineered JSON protocol for Elgato light strips and built automated controller. Has been continuously running on Raspberry Pi 400 since December 2023.

PUBLICATIONS

- Brown, Chase J., et al. "Visualizing digital architectural data for heritage education." IS&T Conference on Visualization and Data Analysis. 2023. Real-time 3D visualization of the Roman forum using three.js, potree, and lidar scans collected by team members in the Notre Dame School of Architecture.
- Pending publication: Aeolian Patterns in Sand.
- Pending publication: Survey of Particle Simulations.